Week 6: Efficient Market Hypothesis (EMH) and Implications

CAPM: Diagnostics for the Disturbance Term (Continued)

What should we do now?

- We've seen that we can reject both the **null of homoskedasticity** and the **null of no autocorrelation** for the 3-factor CAPM model on technology portfolio returns.

- OLS standard errors for $\alpha_i, \beta_{i1}, \cdots, \beta_{i3}$ are not correct.

- What can we do in such cases?

- Point estimates are fine, need to adjust the standard errors.

- This can be done through the use of standard errors that are robust in the presence of heteroskedasticity and autocorrelation.

Newey & West Standard Errors

- Newey and West (1987) propose a covariance estimator that is consistent in the presence of both heteroskedasticity and autocorrelation (HAC) of unknown form.
- These standard errors can be calculated in R easily.
- Let's see how to do this in the case of the 3-factor CAPM for the tech. portfolio.

HAC

```r
ff3f.fit <- lm(data.xts$TECH_RF~data.xts$MKT_RF
                +data.xts$SMB+data.xts$HML)
summary(ff3f.fit)

library(lmtest)
library(sandwich)

T <- nobs(ff3f.fit)
m <- floor(0.75 * T^(1/3))

# compute newey-west HAC uses library(sandwich)
NW_VCOV <- NeweyWest(ff3f.fit,
                     lag = m-1 , prewhite = F,
                     adjust = T)

# Std errors, t-test, p-value uses library(lmtest)
coeftest(ff3f.fit, vcov = NW_VCOV)
```

```
t test of coefficients:

                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.200697   0.103077  1.9471  0.05178 .
data.xts$MKT_RF  1.212908   0.031616 38.3637  < 2e-16 ***
data.xts$SMB     0.947409   0.077403 12.2399  < 2e-16 ***
data.xts$HML    -0.188331   0.072819 -2.5863  0.00983 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Exchange Traded Funds

At the end of June each year, Fama and French sort eligible stocks into:

Size: Small (S) or Big (B), using the NYSE median market equity

Book-to-market: Low (L), Neutral (N), or High (H), using the 30th and 70th NYSE percentiles of BE/ME

This creates 6 portfolios.

	Book-to-market (BtM)		
	Low	Neutral	High
Small	S/L	S/N	S/H
Big	B/L	B/N	B/H

SMB

SMB = Small Minus Big

Take the average return on the three small-stock portfolios and subtract the average return on the three big-stock portfolios:

$$\text{SMB} = \frac{1}{3}(S/L + S/N + S/H) - \frac{1}{3}(B/L + B/N + B/H)$$

So SMB asks:

On average, did small firms outperform big firms this period?

HML

HML = High Minus Low

Take the average return on the two high book-to-market portfolios and subtract the average return on the two low book-to-market portfolios:

$$\text{HML} = \frac{1}{2}(S/H + B/H) - \frac{1}{2}(S/L + B/L)$$

So HML asks:

Did value stocks (high BE/ME) outperform growth stocks (low BE/ME)?

Quant (factor) investing

Such factor models have also huge practical applications.

Many ETFs target these factors directly (Quant (factor) investing).

Moneysmart .gov.au

Exchange traded funds (ETFs)

Exchange traded funds (ETFs) can be a low-cost way to earn a return similar to an index or a commodity.

Vanguard Investments Superannuation Adviser solutions

Do you not like money?

Start investing in future you with Vanguard's low-cost investments and low-fee super.

VUG Vanguard Growth ETF

Overview | **Performance & fees** | **Price** | **Portfolio composition**

Key facts

IOV ticker symbol	VUG.IV
CUSIP	922908736
Management style	Index
Asset class	Domestic Stock - General
Category	Large Growth
Inception date	01/26/2004

VV Vanguard Large-Cap ETF

Overview | Performance & fees | Price | Portfolio compositio

Key facts

IOV ticker symbol	VV.IV
CUSIP	922908637
Management style	Index
Asset class	Domestic Stock - General
Category	Large Blend
Inception date	01/27/2004

For US-listed Vanguard ETFs:

- **Large growth:** VUG
- **Large blend/core:** VV
- **Large value:** VTV
- **Mid growth:** VOT
- **Mid blend/core:** VO
- **Mid value:** VOE
- **Small growth:** VBK
- **Small blend/core:** VB
- **Small value:** VBR

Week 3 Workshop: This is the classic Fama table.

		BtM				
		Low	**2**	**3**	**4**	**high**
	small	0.30	0.81	0.85	1.03	1.17
	2	0.48	0.73	0.93	0.96	1.04
size	**3**	0.50	0.79	0.79	0.89	1.09
	4	0.60	0.57	0.72	0.87	0.87
	Big	0.46	0.52	0.49	0.57	0.65

Drawing on the Fama–French framework, let's consider the Vanguard style-box table - why not?

	Growth	Blend	Value
Small	VBK	VB	VBR
Mid	VOT	VO	VOE
Large	VUG	VV	VTV

ProShares

Our ETFs Strategies Insights

SDS

UltraShort S&P500

Why SDS?

Seek to Profit from Downturns

SDS is the only -2x ETF designed to profit when the daily price of the S&P 500 declines.

Portfolio Hedging

SDS provides the opportunity to offset an expected market decline.

SDS is an ETF

Why SDS?

Seek to Profit from Downturns

SDS is the only -2x ETF designed to profit when the daily price of the S&P 500 declines.

ProShares claims that SDS (ticker symbol: SDS) aims to provide approximately -2 times the daily return of the S&P 500 Index (`^GSPC`).

Using the concepts learned in this unit, explain how you would evaluate whether ProShares UltraShort S&P500 (SDS) appropriately tracks its stated daily target.

To do this, download daily adjusted closing price data for SDS and daily closing price data for the S&P 500 Index (`^GSPC`) from Yahoo Finance, using a sample period that starts on 1 July 2006.

Efficient Market Hypothesis

Motivation

1. The goal of investing is to earn a profit.
2. The most meaningful measure of profit for investors is returns.
3. *Expectations* of returns are a key consideration. They affect asset demand and the price of securities.
4. A contentious topic in modern finance is whether or not returns are predictable:
    - Not the case if all available information about an asset is automatically accounted for in its price, and thereby its return.
    - If returns are predictable, it would be possible to obtain a positive return on some asset without any risk.
    - You could, in a sense, “beat” the market.

Motivation

- The idea that the market can not be “beaten” is sometimes referred to as the **Efficient Markets Hypothesis** (EMH).
- **Fama**’s 1965 and especially 1970 papers made EMH the standard framework in modern finance.

EMH

If true, the EMH implies that:

1. The current price of an asset reflects all information available in the market.
2. The current price provides no information regarding future asset movements.
3. Future returns are completely unpredictable, given information on past returns.
4. Traders can not systematically use newly arriving information to make a profit.
5. Conditional on all previous information, returns are completely random.

Forms of EMH

- What are the forms of EMH?
- The **weak form** (or random walk theory): prices reflect all the information in past prices => future changes in stock prices should be unpredictable.
- The **semi-strong form**: prices reflect all publicly available information => if information is publicly available, then a positive announcement about a company will not, on average, raise the price of its stock because this information has already been *priced in.*
- The **strong form**: prices reflect all acquirable information.

Testing EMH

To understand whether EMH is satisfied, we must consider a broad econometric framework that allows us to test for satisfaction of the EMH.

Let y_t be our variable of interest (this could be returns, but need not necessarily be returns).

Define $\mathcal{F}_{t-1}$ to be all information available to the econometrician at time $t-1$.

This could consist of possible lagged values of y_t, such as $y_{t-1}, y_{t-2}, ..., y_1$, or additional variables such as covariates.

To test the EMH, we will use time series models and methods.

A time series $\{\epsilon_t\}_{t\geq 1}$ is called a **white noise** process if

1. $\mathbb{E}[\epsilon_t] = 0$;

2. $\mathbb{V}[\epsilon_t] = \sigma_\epsilon^2$ for all t;

3. $\gamma_\epsilon(k) = \mathrm{Corr}(\epsilon_t, \epsilon_{t-k}) = 0$, for all t and k.

This is sometimes written as $\epsilon_t \sim \mathrm{WN}(0, \sigma_\epsilon^2)$.

A model for log returns

Suppose

$$r_t = \mu + \epsilon_t,$$

and $\epsilon_t \sim \mathrm{WN}(0, \sigma_\epsilon^2)$, then the EMH would be satisfied.

Even if we knew ϵ_t, it would not help us predict, on average, the future values of r_{t+h}, $h \geq 1$.

If we assume the term ϵ_t is white noise, we can test the EMH hypothesis in two ways:

Method 1: Return predictability

If markets are efficient then returns, at any frequency, should not be predictable.

Method 2: To compare the variance of asset returns over different time horizons.

If markets are efficient, then the variance of asset returns over different time horizons should roughly increase with their horizon. If this is the case, the variance of n period returns should simply be n times the variance of the 1-period return.

Measure return predictability through serial dependence.The most common measure of **sample serial dependence** is (sample) autocorrelation function (ACF).

The sample "autocorrelation function" is given by:

$$\hat{\gamma}(k) = \frac{\sum_{t=k+1}^{T}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}.$$

The population ACF is

$$\gamma(k) = \frac{\mathrm{Cov}(r_t, r_{t-k})}{\mathbb{V}(r_t)}.$$

Autocorrelation function (ACF)

1. $\hat{\gamma}(k)$ measures the strength of association (correlation) between returns at time t and returns at time $t - k$.

2. Numerator of $\hat{\gamma}(k)$ is just the covariance of r_t and r_{t-k}.

3. Denominator is a normalization to ensure that $\hat{\gamma}(k)$ is between -1 and 1.

Return Predictability

1. If returns exhibit no predictability, then there should be no autocorrelation, as returns at any period t would not have any correlation with returns at period $t-k$.

2. If returns are not predictable, $\hat{\gamma}(k) \approx 0$ for any $k = 1, 2, ...$

3. If returns exhibit positive or negative autocorrelation, then this pattern can be exploited to predict future behavior, and hence the EMH would not be satisfied.

Examples

Example 1: Consider the hourly observations for the period 00:00 1 January 1986 to 11:00 15 July 1986 on the $ / £ (ie. U.S. dollars per pound, how many US dollars are needed to buy one British pound) exchange rate.

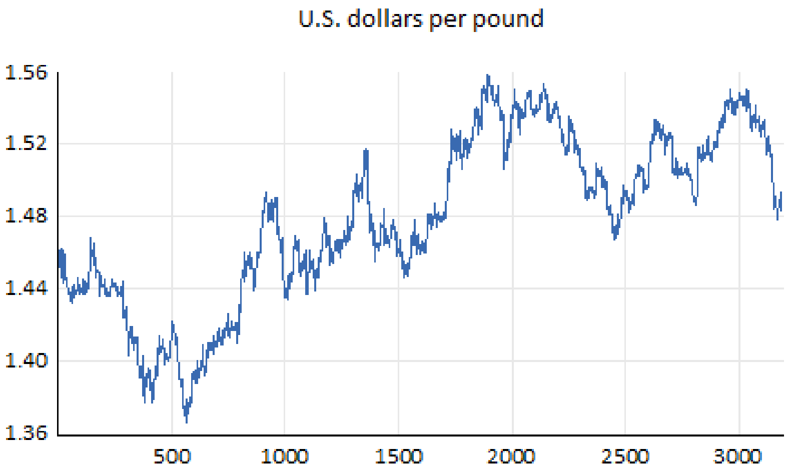
Example 2: Consider the daily prices for the U.S. Apple stock from 14 March 2005 to 13 March 2014.

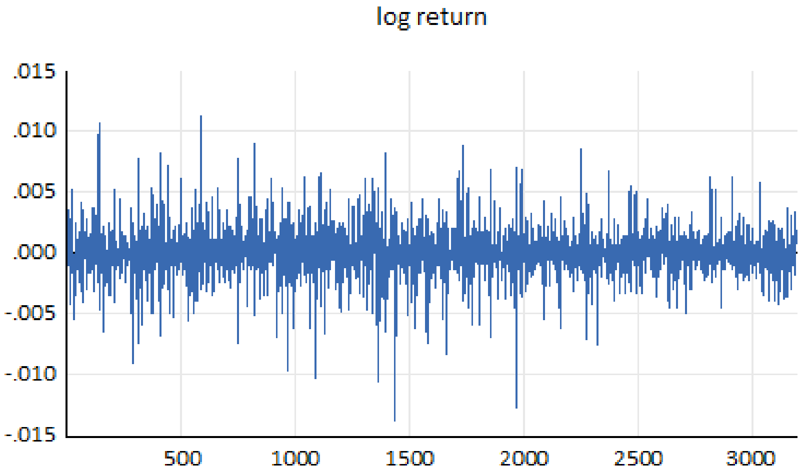
U.S. dollars per pound

```
# plot U.S. dollars per pound
plot(usd_bp.xts)

# log return
logret <- na.omit(diff(log(usd_bp.xts)))

# plot log return
plot(logret, main = "log return")
```


U.S. dollars per pound
1.56
1.52
1.48
1.44
1.40
1.36
500
1000
1500
2000
2500
3000

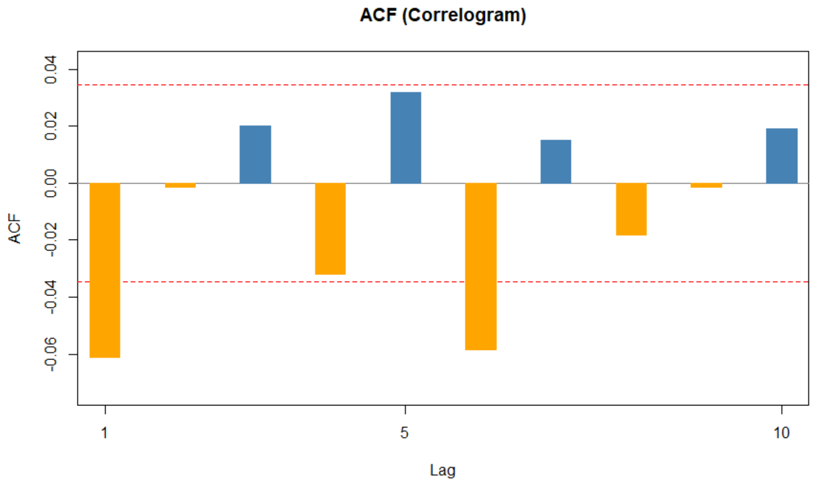

log return
.015
.010
.005
.000
-.005
-.010
-.015
500
1000
1500
2000
2500
3000

```
> source("Correlogram.R")

> length(logret)
[1] 3189

> correlogram(logret,max.lag = 10)
 Lag        AC       Q      Prob
   1 -0.0610 11.8834 5.664e-04
   2 -0.0013 11.8891 2.620e-03
   3  0.0202 13.1934 4.237e-03
   4 -0.0318 16.4287 2.495e-03
   5  0.0318 19.6576 1.449e-03
   6 -0.0584 30.5479 3.092e-05
   7  0.0152 31.2856 5.507e-05
   8 -0.0182 32.3426 8.086e-05
   9 -0.0013 32.3479 1.733e-04
  10  0.0190 33.4976 2.246e-04
```

Correlogram



Correlogram

The “Q” column represents a joint test of the significance of the autocorrelations. Specifically the k^{th} Q statistic is a test of the null hypothesis

$$H_0:\ \gamma_1 = \gamma_2 = \cdots = \gamma_k = 0,$$

against

$$H_1:\ \text{at least one of } \gamma_1, \ldots, \gamma_k \neq 0.$$

The last column heading “Prob” contains the p values for these Q statistics.

If the p value is less than 0.05 then we reject H_0 and conclude that there is at least one significant autocorrelation at lag k or less.

Correlogram

The dashed lines on the graph represent the 95% confidence interval of the autocorrelation, which is equal to

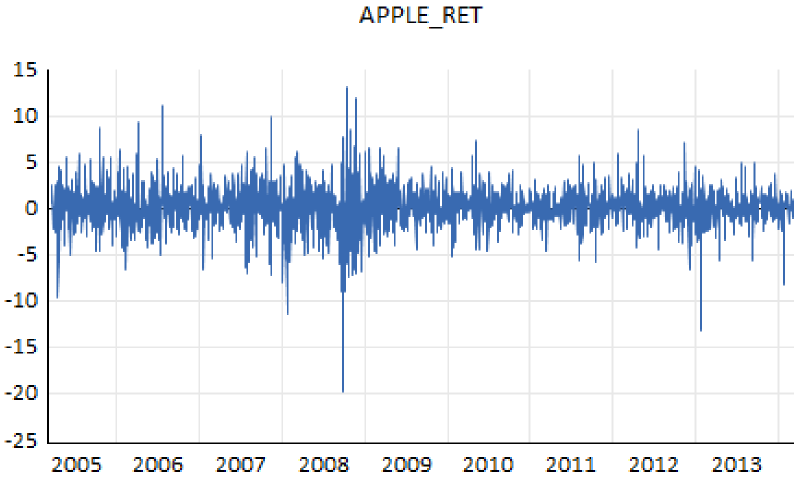
$$\pm \frac{2}{\sqrt{T}}$$

Any autocorrelation which goes outside of the dotted lines to either side is significantly different from zero.

```
> # plot Apple
> plot(prices.xts)

> # log return
> lr <- na.omit(diff(log(prices.xts)))*100
```


APPLE
700
600
500
400
300
200
100
0
2005
2006
2007
2008
2009
2010
2011
2012
2013

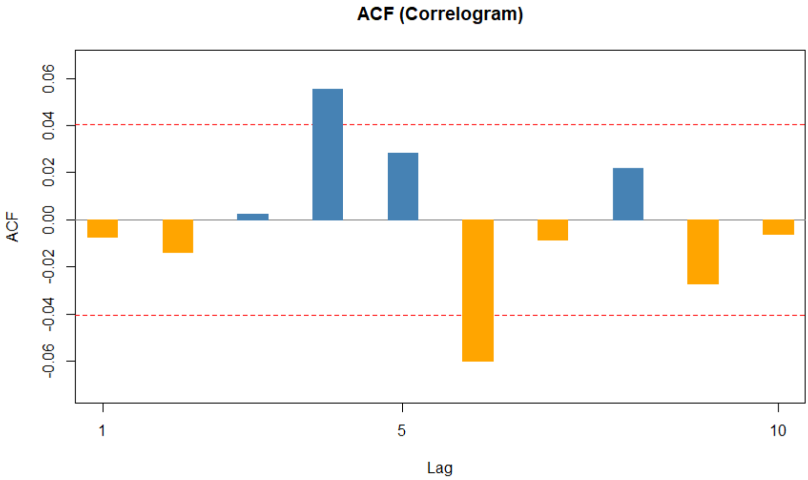

APPLE_RET
15
10
5
0
-5
-10
-15
-20
-25
2005
2006
2007
2008
2009
2010
2011
2012
2013

Correlogram

```
> source("correlogram.R")

> length(lr)
[1] 2348

> correlogram(lr,max.lag = 10)
 Lag      AC       Q     Prob
   1 -0.0072  0.1230 0.725800
   2 -0.0137  0.5619 0.755100
   3  0.0025  0.5770 0.901700
   4  0.0553  7.7662 0.100500
   5  0.0284  9.6641 0.085330
   6 -0.0600 18.1523 0.005863
   7 -0.0085 18.3245 0.010590
   8  0.0218 19.4402 0.012670
   9 -0.0271 21.1726 0.011910
  10 -0.0061 21.2604 0.019350
```


ACF (Correlogram)
ACF
0.06
0.04
0.02
0.00
-0.02
-0.04
-0.06
1
5
10
Lag

The correlogram under the column AC gives **sample autocorrelations**, of various lag lengths, for the log return of $/£ and Apple.

It is clear that all autocorrelations are close to zero. This suggests that these stocks are not predictable using their own past histories.

While the ACFs for returns look small, to be certain of the validity of the EMH, we test their significance.

Hypothesis Test:

- *If $\{y_t\}$ is stationary, when the sample size T is large, $\hat{\gamma}(k)$ should be approximately normal.*

- $\hat{\gamma}(k) \sim N(0, 1/T)$*, so that* $\sqrt{T}\hat{\gamma}(k)$ *will be asymptotically standard normal.*

- *Test of EMH*

$$H_0 : \gamma(k) = 0$$
$$H_1 : \gamma(k) \neq 0$$

- *Decision rule: reject H_0 if $|\sqrt{T}\hat{\gamma}(k)|$ is larger than 1.96, which is the critical value for the standard normal associated with a two-sided hypothesis test at the 5% significance level.*

Test of EMH

1. This hypothesis test means that a very simple statistical test of the EMH exists.
2. If $\hat{\gamma}(k)$ is statistically different from zero $\implies$ EMH cannot be satisfied.

The EMH can be tested simply by testing the most important ACF, namely, $\hat{\gamma}(1)$.

From the correlogram, we have

\$/£: $\hat{\gamma}(1) = -0.061$, Test statistic = $\sqrt{3189} * -0.061 = -3.44$

Apple: $\hat{\gamma}(1) = -0.007$, Test statistic = $\sqrt{2348} * -0.007 = -0.34$

We cannot reject the EMH for Apple but we can for \$/£. This implies that \$/£ returns are predictable.

Testing for Autocorrelation

1. We can use the LM test for autocorrelation in our analysis of the CAPM.

2. To construct such an LM test, recall the idea of an auxiliary regression model. In this case, we have the residuals being

$$\hat{\epsilon}_t = r_t - \bar{r}, \quad \bar{r} = \sum_{t=1}^{T} r_t / T.$$

3. Test for autocorrelation of $\hat{\epsilon}_t$ using the auxiliary regression model

$$\hat{\epsilon}_t = \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1} + \gamma_2 \hat{\epsilon}_{t-2} + \cdots + \gamma_k \hat{\epsilon}_{t-k} + v_t$$

where v_t is some random disturbance term.

- Test

$$H_0 : \gamma_1 = \cdots = \gamma_k = 0 \text{ [EMH is satisfied]}$$
$$H_1 : \text{at least one differs from zero [EMH is } \textbf{not} \text{ satisfied]}$$

- We can use as our test statistic $AR(k) = T \cdot R^2$ which will have a χ^2_k distribution under the null hypothesis.
- We can compare this with the corresponding χ^2_k critical value to assess whether or not we reject the null hypothesis.

Example

Let us test for serial correlation using returns on \$ / £ .

- The null and alternative hypotheses are given by:

$$
\begin{aligned}
H_0 :& \gamma_1 = \gamma_2 = 0 \\
H_1 :& \text{ At least one differs from zero}
\end{aligned}
$$

- From the auxiliary regression model:

$$\hat{\epsilon}_t = \gamma_0 + \gamma_1 \hat{\epsilon}_{t-1} + \gamma_2 \hat{\epsilon}_{t-2} + v_t.$$

- The test statistic $\mathrm{AR}(2) = T \cdot R^2 = 3189 * 0.003749 = 11.955$
- We will reject the null hypothesis at the 5% significance level if $\mathrm{AR}(2) > \chi_2^2(.05)$
- We can reject the null of EMH!

```
> fit <- lm(logret ~ 1)

> summary(fit)

Call:
lm(formula = logret ~ 1)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0139146 -0.0007195 -0.0000090  0.0007195  0.0111844

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 9.05e-06   3.32e-05   0.273    0.785

Residual standard error: 0.001875 on 3188 degrees of freedom
```

BG-LM test

```
> library(lmtest)

> bgtest(fit, order = 2, type = "Chisq")

        Breusch-Godfrey test for serial correlation of order up

data:  fit
LM test = 11.955, df = 2, p-value = 0.002535
```

Variance ratio

An alternative way to examine the efficient markets hypothesis is to compare the variance on returns over different time horizons.

Recall the 1-period log returns

$$r_t = \log P_t - \log P_{t-1}$$

and the n-period returns

$$r_{nt} = \log P_t - \log P_{t-n} = r_t + r_{t-1} + \cdots + r_{t-(n-1)},$$

Consider the variances of the 1-period returns and the n-period returns

$$s_1^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2, \qquad s_n^2 = \frac{1}{T}\sum_{t=1}^{T}(r_{nt} - n\bar{r})^2,$$

where $n\bar{r}$ represents the sample mean of n-period returns r_{nt}.

If there is no autocorrelation, the variance of n-period returns should equal n times the variance of the 1-period returns.

The ratio

$$VR_n = \frac{s_n^2}{n s_1^2},$$

is known as the variance ratio.

Variance ratio

The variance ratio has the following implications for the properties of log returns:

$$VR_n = \begin{cases} = 1 & \text{[No autocorrelation]} \\ > 1 & \text{[Positive autocorrelation]} \\ < 1 & \text{[Negative autocorrelation]} \end{cases}$$

Consider an $n = 3$ period return

$$r_{3t} = r_t + r_{t-1} + r_{t-2},$$

which is the sum of the three 1-period returns.

Let the sample mean for the 1-period returns be $\bar{r}$.

Subtracting $\bar{r}$ from both sides three times gives

$$(r_{3t} - 3\bar{r}) = (r_t - \bar{r}) + (r_{t-1} - \bar{r}) + (r_{t-2} - \bar{r}).$$

By squaring both sides and averaging over a sample of size T, we obtain the expression shown in the next slide.

$$\frac{1}{T}\sum_{t=1}^{T}(r_{3t}-3\bar{r})^2$$

$$=\frac{1}{T}\sum_{t=1}^{T}(r_t-\bar{r})^2+\frac{1}{T}\sum_{t=1}^{T}(r_{t-1}-\bar{r})^2+\frac{1}{T}\sum_{t=1}^{T}(r_{t-2}-\bar{r})^2 \text{ [Var of } r_t, r_{t-1}, r_{t-2}\text{]}$$

$$+\frac{2}{T}\sum_{t=1}^{T}(r_t-\bar{r})(r_{t-1}-\bar{r}) \quad \text{[Autocovariance of } r_t, r_{t-1}\text{]}$$

$$+\frac{2}{T}\sum_{t=1}^{T}(r_t-\bar{r})(r_{t-2}-\bar{r}) \quad \text{[Autocovariance of } r_t, r_{t-2}\text{]}$$

$$+\frac{2}{T}\sum_{t=1}^{T}(r_{t-1}-\bar{r})(r_{t-2}-\bar{r}) \quad \text{[Autocovariance of } r_{t-1}, r_{t-2}\text{]}$$

Variance ratio

This expansion requires values for r_0 and r_{-1}. To implement this formulation in practice, the summation ranges are suitably adjusted to the available data.

In the case of zero autocovariances (no autocorrelation), the relationship simplifies to

$$s_3^2 = \frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2 + \frac{1}{T}\sum_{t=1}^{T}(r_{t-1} - \bar{r})^2 + \frac{1}{T}\sum_{t=1}^{T}(r_{t-2} - \bar{r})^2.$$

Variance ratio

Assuming that the variance for r_t is the same as the variance for r_{t-1} and r_{t-2}, then in the case of no autocorrelation in $n = 3$ period returns

$$s_3^2 = 3s_1^2.$$

The assumption of equal variance for r_{t-1} and r_{t-2} is known as stationarity.

Stationarity

The above measures can only be computed in a meaningful way if the time series is **stationary**.

Definition

A time series y_t is **(weakly or covariance) stationary** if for all t

1. $\mathbb{E}[y_t] = \mu$;
2. $\mathbb{V}[y_t] = \sigma^2$;
3. $\gamma(j)$ depends only on j but not on t, for $j = 1, 2, \ldots$.

```r
# Suppose r.xts is your xts return series
r <- as.numeric(na.omit(r.xts))
n <- length(r)

# Overlapping 3-period returns
r3 <- r[1:(n-2)] + r[2:(n-1)] + r[3:n]

# Simple variance ratio for k = 3
VR3 <- var(r3) / (3 * var(r))

VR3
```

Using returns on $ / £

```
> r <- logret

> n <- length(r)

> # Overlapping 3-period returns
> r3 <- r[1:(n-2)] + r[2:(n-1)] + r[3:n]

> # Simple variance ratio for k = 3
> VR3 <- var(r3) / (3 * var(r))

> VR3
          bp
bp 3.000257
```